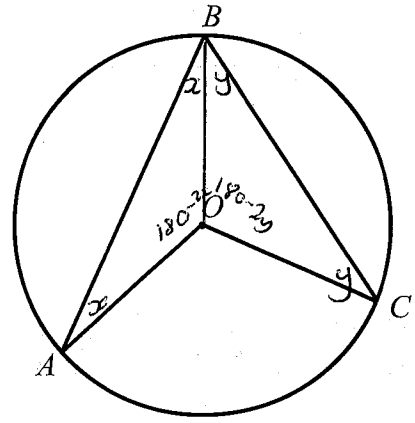
**1**




Let $ABO = x$

Let $CBO = y$

*A*, *B* and *C* are points on the circumference of a circle, centre *O*.

Prove that angle *AOC* is twice the size of angle *ABC*.
You must **not** use any circle theorems in your proof.

$ABO = BAO$ Angles at the base of an isosceles triangle are equal

$CBO = BCO$ ——— " ———

$AOB = 180 - 2x$ Angles in a triangle add to $180°$

$BOC = 180 - 2y$ ——— " ———

$ABC = x + y$

$AOC = 360 - (180 - 2x) - (180 - 2y)$

$= 360 - 180 + 2x - 180 + 2y$

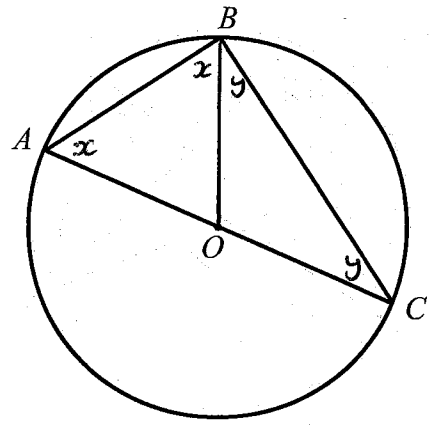
$= 2x + 2y$

$= 2(x + y)$

$= 2(ABC)$

**(Total for Question 1 is 4 marks)**

**2**




Let $OAB = x$

Let $OBC = y$

*A*, *B* and *C* are points on the circumference of a circle, centre *O*.
*AOC* is a diameter of the circle.

Prove that angle *ABC* is 90°
You must **not** use any circle theorems in your proof.

$OAB = OBA$ Angles at the base of an isosceles triangle are equal

$OBC = OCB$ ——— " ———

$AOB = 180 - 2x$
$BOC = 180 - 2y$
Angles in a triangle add to 180°

$180 - 2x + 180 - 2y = 180$

$360 - 2x - 2y = 180$

$180 = 2x + 2y$

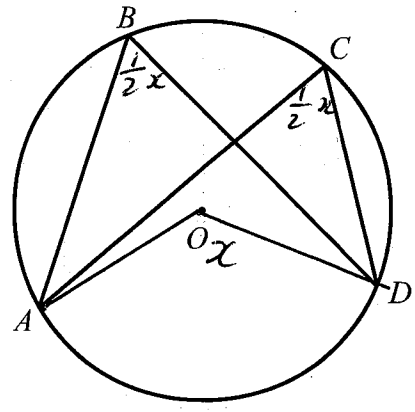
$90 = x + y$

$ABC = x + y$

$= 90°$

**(Total for Question 2 is 4 marks)**

**3**




*A*, *B*, *C* and *D* are points on the circumference of a circle, centre *O*.

Prove that angle *ABD* and angle *ACD* are equal.

$AOD = x$

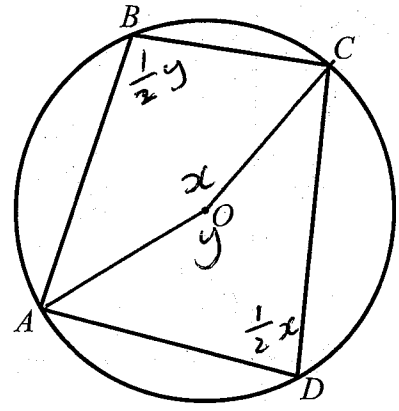
$ABD = \frac{1}{2}x$ Angle at the circumference is half the angle at the centre

$ACD = \frac{1}{2}x$ ——— " ———

$ABD = ACD = \frac{1}{2}x$

**(Total for Question 3 is 2 marks)**

**4**




*A*, *B*, *C* and *D* are points on the circumference of a circle, centre *O*.

Prove that angle *ABC* and angle *ADC* add to 180°

Let AOC (minor) $= x$

Let AOC (major) $= y$

$ACD = \frac{1}{2}x$ The angle at the centre is twice the angle at the circumference.

$ABC = \frac{1}{2}y$ ——— " ———

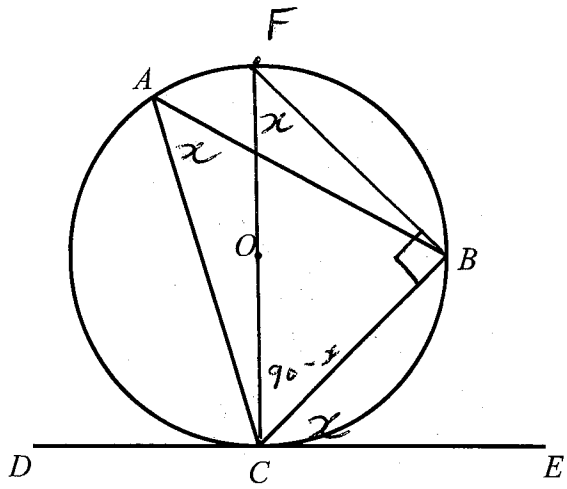
$x + y = 360$ Angles around a point add to 360°

$\therefore \frac{1}{2}x + \frac{1}{2}y = 180$

$ABC + ADC = 180$

**(Total for Question 4 is 4 marks)**

**5**




*A*, *B* and *C* are points on the circumference of a circle, centre *O*.
*DCE* is a tangent to the circle.

Prove that angle *BCE* and angle *BAC* are equal.

Let $BCE = x$

$OCE = 90°$ Tangent meets radius at $90°$

$OCB = 90 - x$

$FBC = 90°$ Angle in a semi-circle is $90°$

$BFC = 180 - 90 - (90 - x)$

$= x$

$BAC = x$

~~$BCE = x$ $BFC = x$~~

Angles from the same points (in same segment) are equal.

$BAC = BCE = x$

**(Total for Question 5 is 4 marks)**